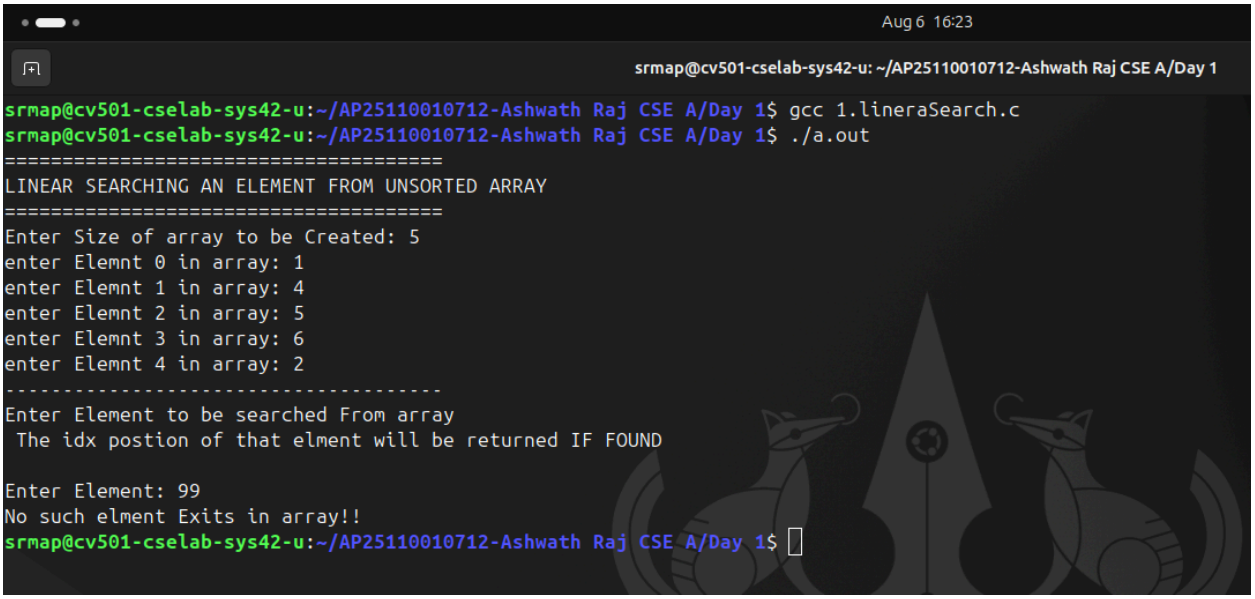
Ashwath Raj M| AP25110010712| CSE core A

GITHUB LINK: *https://github.com/Ashwath-Raj/CSE-204-DAA-assignments-*

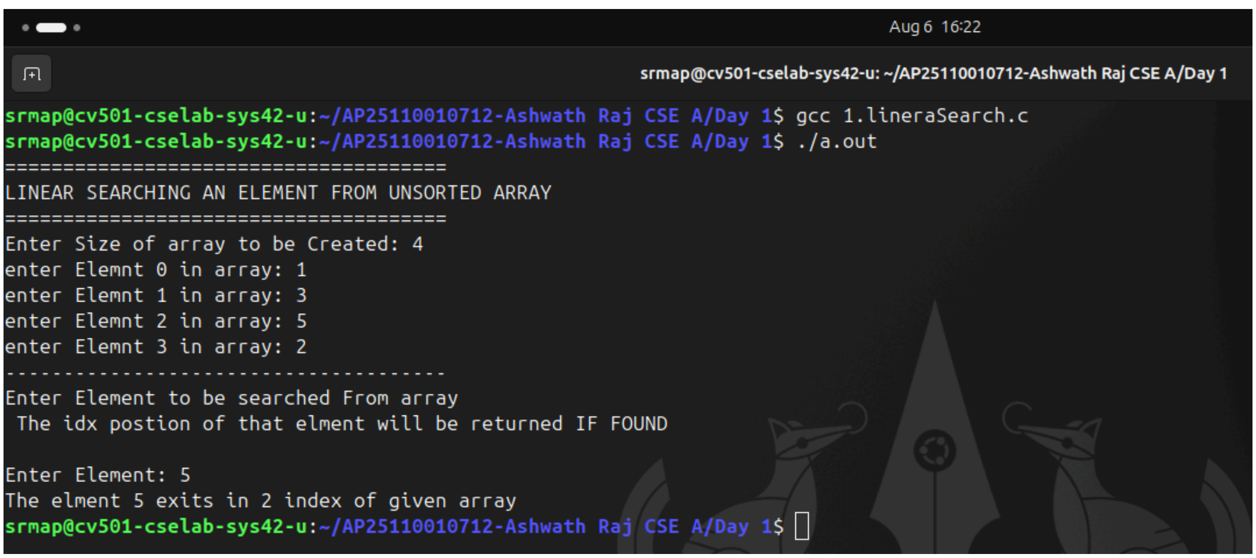
Linear Search

Output:



```
Aug 6  16:23
srmap@cv501-cselab-sys42-u: ~/AP25110010712-Ashwath Raj CSE A/Day 1
srmap@cv501-cselab-sys42-u:~/AP25110010712-Ashwath Raj CSE A/Day 1$ gcc 1.lineraSearch.c
srmap@cv501-cselab-sys42-u:~/AP25110010712-Ashwath Raj CSE A/Day 1$ ./a.out
======================================
LINEAR SEARCHING AN ELEMENT FROM UNSORTED ARRAY
======================================
Enter Size of array to be Created: 5
enter Elemnt 0 in array: 1
enter Elemnt 1 in array: 4
enter Elemnt 2 in array: 5
enter Elemnt 3 in array: 6
enter Elemnt 4 in array: 2
--------------------------------------
Enter Element to be searched From array
 The idx postion of that elment will be returned IF FOUND

Enter Element: 99
No such elment Exits in array!!
srmap@cv501-cselab-sys42-u:~/AP25110010712-Ashwath Raj CSE A/Day 1$
```



```
Aug 6  16:22
srmap@cv501-cselab-sys42-u: ~/AP25110010712-Ashwath Raj CSE A/Day 1
srmap@cv501-cselab-sys42-u:~/AP25110010712-Ashwath Raj CSE A/Day 1$ gcc 1.lineraSearch.c
srmap@cv501-cselab-sys42-u:~/AP25110010712-Ashwath Raj CSE A/Day 1$ ./a.out
======================================
LINEAR SEARCHING AN ELEMENT FROM UNSORTED ARRAY
======================================
Enter Size of array to be Created: 4
enter Elemnt 0 in array: 1
enter Elemnt 1 in array: 3
enter Elemnt 2 in array: 5
enter Elemnt 3 in array: 2
--------------------------------------
Enter Element to be searched From array
 The idx postion of that elment will be returned IF FOUND

Enter Element: 5
The elment 5 exits in 2 index of given array
srmap@cv501-cselab-sys42-u:~/AP25110010712-Ashwath Raj CSE A/Day 1$
```

Code:

```
/*
Program Name: LInear Search Implememtation
Program Description: Takes THe inputs : {arr, arr size} intializes an
Array and Uses LInear Seach Algorithm to find desired number
            From Array IF it exists
Proram CReation Time: Aug 6th 15:40
Program Creater: Ashwath Raj | AP25110010712
*/

#include<stdio.h>

int linearSearch(int arr[], int n, int key);

void main(void) {
   printf("======================================= \n");
   printf("LINEAR SEARCHING AN ELEMENT FROM UNSORTED ARRAY \n");
   printf("======================================= \n");


   int n = 0;
   printf("Enter Size of array to be Created: ");
   scanf("%d", &n);

   int arr[n];
   for (int i = 0; i < n; i++) {
       printf("enter Elemnt %d in array: ", i);
       scanf("%d", &arr[i]);
   }

   int key = -1;
   printf("--------------------------------------- \n");
   printf("Enter Element to be searched From array \n The idx postion of
that elment will be returned IF FOUND \n \n");
   printf("Enter Element: ");
   getchar();
   scanf("%d", &key);
```

```
    int idx =  linearSearch(arr, n, key);
    if (idx == -1) {
        printf("No such elment Exits in array!! \n");
    }
    else {
        printf("The elment %d exits in %d index of given array \n", key,
idx);
    }
}

int linearSearch(int arr[],int n,  int key) {
    for(int i = 0; i < n; i++) {
        if(arr[i] == key) {
            return i;
        }
    }

    return -1;
}
```

Linear Search

Output:

```
Aug 6  16:24
srmap@cv501-cselab-sys42-u: ~/AP25110010712-Ashwath Raj CSE A/Day 1
srmap@cv501-cselab-sys42-u:~/AP25110010712-Ashwath Raj CSE A/Day 1$ gcc 2.BinarySearch.c
srmap@cv501-cselab-sys42-u:~/AP25110010712-Ashwath Raj CSE A/Day 1$ ./a.out
======================================
Binary SEARCHING AN ELEMENT FROM UNSORTED ARRAY
======================================
Enter Size of array to be Created: 4
enter Element 0 in array: 2
enter Element 1 in array: 3
enter Element 2 in array: 6
enter Element 3 in array: 8
--------------------------------------
Enter Element to be searched From array
 A validation statement will be displayed IF element Exists

Enter Element: 2
The element 2 exits in given array
srmap@cv501-cselab-sys42-u:~/AP25110010712-Ashwath Raj CSE A/Day 1$ 
```

```
Aug 6  16:25
srmap@cv501-cselab-sys42-u: ~/AP25110010712-Ashwath Raj CSE A/Day 1
srmap@cv501-cselab-sys42-u:~/AP25110010712-Ashwath Raj CSE A/Day 1$ gcc 2.BinarySearch.c
srmap@cv501-cselab-sys42-u:~/AP25110010712-Ashwath Raj CSE A/Day 1$ ./a.out
======================================
Binary SEARCHING AN ELEMENT FROM UNSORTED ARRAY
======================================
Enter Size of array to be Created: 5
enter Element 0 in array: 1
enter Element 1 in array: 6
enter Element 2 in array: 8
enter Element 3 in array: 9
enter Element 4 in array: 11
--------------------------------------
Enter Element to be searched From array
 A validation statement will be displayed IF element Exists

Enter Element: 14
No such element Exits in array!!
srmap@cv501-cselab-sys42-u:~/AP25110010712-Ashwath Raj CSE A/Day 1$ 
```

Code:

```
/*
Program Name: Binary Search Implementation
Program Description: Takes The inputs : {arr, arr size} initializes an
Array and Uses Binary Search Algorithm to find desired number
            From Array IF it exists
Program Creation Time: Aug 6th 16:00
Program Creator: Ashwath Raj | AP25110010712
*/

#include<stdio.h>

void swap(int* a, int* b);
void sortAnArray(int arr[], int n);
int binarySearch(int arr[], int n, int key);

void main(void) {
   printf("======================================= \n");
   printf("Binary SEARCHING AN ELEMENT FROM UNSORTED ARRAY \n");
   printf("======================================= \n");


   int n = 0;
   printf("Enter Size of array to be Created: ");
   scanf("%d", &n);

   int arr[n];
   for (int i = 0; i < n; i++) {
     printf("enter Element %d in array: ", i);
     scanf("%d", &arr[i]);
   }

   int key = -1;
   printf("--------------------------------------- \n");
   printf("Enter Element to be searched From array \n A validation
statement will be displayed IF element Exists \n \n");
   printf("Enter Element: ");
   getchar();
```

```c
   scanf("%d", &key);

       sortAnArray(arr, n);
   int found =  binarySearch(arr, n, key);
   if (found == -1) {
       printf("No such element Exits in array!! \n");
   }
   else {
       printf("The element %d exits in given array \n", key);
   }
}

void swap(int* a, int* b) {
 int temp = *a;
 *a = *b;
 *b = temp;
}

void sortAnArray(int arr[], int n) {
 for(int i = 0; i < n; i++) {
   for(int j = 1; j < n; j++) {
     if(arr[j] < arr[j - 1])
       swap(&arr[j], &arr[j - 1]);
   }
 }
}

int binarySearch(int arr[], int n, int key) {
 int left = 0,right = n - 1;
  while (left <= right) {
   int mid = left + (right - left) / 2;
   if (arr[mid] == key)
     return mid;
   else if(arr[mid] > key)
     right = mid - 1;
   else if (arr[mid] < key)
     left  = mid + 1;
 }
 return -1;
}
```